We have computed the Lie bracket for $\mathfrak{sl}(2,\mathbb{C})$ using the vector field commutator.

Define $X_i = \left(\frac{\partial}{\partial x^i}\right)_{id}$

$$[X_1, X_2] = 2X_2 \;;\; [X_2, X_3] = X_1 \;;\; [X_3, X_1] = 2X_3$$

General

$$[X_i, X_j] = C^k{}_{ij} X_k \Rightarrow C^2{}_{12} = 2 = C^3{}_{31} \;;\; C^1{}_{23} = 1.$$

- Killing form of $\mathfrak{sl}(2,\mathbb{C})$.

$$K(A,B) = \mathrm{tr}(\mathrm{ad}(A) \circ \mathrm{ad}(B))$$

$$K(X_i, X_j) = K_{ij} = C^m{}_{in} C^n{}_{jm} \quad \text{symmetric matrix.}$$

$$= C^n{}_{jm} C^m{}_{in}$$

$$= C^m{}_{jn} C^n{}_{im} = K_{ji}$$

$$\odot\; K_{11} = C^m{}_{1n} C^n{}_{1m} = C^2{}_{12} C^2{}_{12} + C^3{}_{13} C^3{}_{13} = 8.$$

$$K_{22} = C^m{}_{2n} C^n{}_{2m} = C^m{}_{21} C^1{}_{2m} + C^m{}_{23} C^3{}_{2m} = 0$$

$$K_{33} = C^m{}_{3n} C^n{}_{3m} = 0.$$

$$K_{12} = C^m{}_{1n} C^n{}_{2m} = C^m{}_{11} + C^2{}_{12} C^2{}_{22} + C^3{}_{13} C^3{}_{23} = 0.$$

$$K_{13} = C^m{}_{1n} C^n{}_{3m} = C^3{}_{13} C^3{}_{33} + C^2{}_{12} C^2{}_{32} = 0$$

$$K_{23} = C^m{}_{2n} C^n{}_{3m} = C^2{}_{21} C^1{}_{32} + C^1{}_{23} C^3{}_{31} = 4.$$

$$(K_{ij}) = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}$$

$\circledast$ Recall Cartan Criteria: $(2^{nd}) \to (L, [\cdot,\cdot])$ Lie algebra is semi-simple iff K is non-degenerate $\Leftrightarrow \det(K_{ij}) \neq 0$.

$\circledast$ $\det(K_{ij}) = -128 \neq 0. \Rightarrow \mathfrak{sl}(2,\mathbb{C})$ is semi-simple.

$\Rightarrow L_1 \oplus L_2 \oplus \cdots \oplus L_n$ (each L_i simple)

$\circledast$ If G is a compact Lie group, then K negative definite. $\rightsquigarrow$ Need Haar Measure.

as K is not negative definite $\Rightarrow SL(2,\mathbb{C})$ is not compact.

- Claim: $\mathfrak{sl}(2,\mathbb{C})$ is simple

Recall: L is simple, if it contains no non-trivial ideals.

$I \subseteq L$ (subspace) if $[I, L] \subseteq I$. $\{0\}$ & L are ideals trivial.

$\circledast$ We try to construct an non-trivial ideal explicitly

$$I = \left\{ \alpha X_1 + \beta X_2 + \gamma X_3 \;\middle|\; \begin{array}{l}\alpha,\beta,\gamma \text{ s.t.} \\ \text{we don't have} \\ \text{a trivial ideal}\end{array} \right\} \;;\; (\alpha,\beta,\gamma \in \mathbb{C}) \times$$

$(\alpha = \beta = \gamma = 0)$ ×

Since I is subspace itself, if $[\![I, X_i]\!] \subseteq I$ for all i, we are done

$\Rightarrow$ only need to compute $[\![\alpha X_1 + \beta X_2 + \gamma X_3, X_i]\!]$

$\Rightarrow$ $i=1 \Rightarrow [\![\alpha X_1 + \beta X_2 + \gamma X_3, X_1]\!] = -2\beta X_2 + 2\gamma X_3$

$i=2 \Rightarrow [\![\alpha X_1 + \beta X_2 + \gamma X_3, X_2]\!] = 2\alpha X_2 - \gamma X_1$

$i=3 \Rightarrow [\![\alpha X_1 + \beta X_2 + \gamma X_3, X_3]\!] = -2\alpha X_3 + \beta X_1$

If $\alpha \neq 0$, $\beta \neq 0$; $\gamma \neq 0$ we got three linearly independent vectors.

Let $Y \in \mathfrak{sl}(2,\mathbb{C})$, $Y = y^i X_i$

$[\![I, L]\!] \subseteq I \xrightarrow{Y} \boxed{I = L}$ ← we don't want this.

Let's start relaxing the conditions. $\Rightarrow$ atleast one of α, β, γ must be zero.

⊙ $\underline{\alpha = 0}$: $[\![\ ,\ ]\!] \subset \text{span}(\{X_2, X_3\})$

$\Rightarrow \gamma = 0$ & $\beta = 0$ $\Rightarrow$ $I = \{0\}$.

⊙ $\underline{\beta = 0}$: $[\![\ ,\ ]\!] \subset \text{span}(\{X_1, X_3\})$

$\Rightarrow \alpha = 0 \Rightarrow [\![\ ,\ ]\!] \subset \text{span}(\{X_3\}) \Rightarrow \gamma = 0 \Rightarrow I = \{0\}$.

⊙ $\underline{\gamma = 0}$: $[\![\ ,\ ]\!] \subset \text{span}(\{X_1, X_2\})$

$\Rightarrow \alpha = 0 \Rightarrow [\![\ ,\ ]\!] \subset \text{span}(\{X_2\}) \Rightarrow \beta = 0 \Rightarrow I = \{0\}$.

$\rightarrow$ $\nexists$ non-trivial ideal of $\mathfrak{sl}(2,\mathbb{C})$.

$\Rightarrow$ $\mathfrak{sl}(2,\mathbb{C})$ is simple.

- $\underline{\text{Roots}}$ and $\underline{\text{Fundamental roots}}$ of $\mathfrak{sl}(2,\mathbb{C})$.

Start by finding the Cartan subalgebra, H

$\underline{\text{Claim}}$: $H = \text{span}_{\mathbb{C}}\{X_1\}$.

basis extension is obv. $\{X_1, X_2, X_3\}$.

$\text{ad}(X_1)X_2 = [\![X_1, X_2]\!] = 2X_2$

$\text{ad}(X_1)X_3 = [\![X_1, X_3]\!] = -2X_3$

$\Rightarrow$ they are indeed eigenvectors.

$\Rightarrow$ H is Cartan subalgebra.

define $\lambda_2 : H \xrightarrow{\sim} \mathbb{C}$

$\forall h \in H : \exists \xi \in \mathbb{C} : h = \xi X_1$

$\Rightarrow \lambda_2(h) = \xi \lambda_2(X_1)$

$\lambda_2(X_1) = 2$, $\lambda_3(X_1) = -2$ } roots.

and observe $\boxed{\lambda_2 = -\lambda_3}$

$\hookrightarrow$ $\lambda_2, \lambda_3 \in H^*$. $\Phi = \{\lambda_2, \lambda_3\}$.

- $\underline{\text{fundamental roots}}$: $\Pi \subset \Phi$

s.t. (a) Π is linearly indepedent

(b) $\forall \lambda \in \Phi \ \exists n_1, \dots, n_{|\Pi|} \in \mathbb{N}_0 \ \exists \varepsilon \in \{+1, -1\}$:

$\lambda = \varepsilon \sum_{i=1}^{|\Pi|} n_i \pi_i$.

Here: $\Pi = \{\lambda_2\}$.